Lesson 3: Differentiation and Integration

AI/ML Mathematics

Contents

1 Motivation and Overview 2

2 References 2

3 The Derivative 3

4 Differentiation Rules 3

5 Mean Value Theorem and Consequences 5

6 Taylor's Theorem 6

7 Riemann Integration 7

8 Properties of the Integral 8

9 Fundamental Theorem of Calculus 9

10 Exercises 10

11 Summary 10

12 Further Reading 11

1 Motivation and Overview

Every modern deep learning system is trained by *gradient descent*: given a loss function $L(\theta)$, we compute its derivative with respect to the parameters θ and update $\theta \leftarrow \theta - \eta \nabla L(\theta)$. The chain rule, applied layer by layer through the network, is the mathematical engine behind backpropagation. But what guarantees that the derivative exists, that the chain rule is valid, and that the loss decrease can be quantified? The *mean value theorem* gives precise bounds on how much a function changes in terms of its derivative, and *Taylor's theorem* provides the polynomial approximation that justifies gradient descent convergence rates.

On the integration side, expected values $\mathbb{E}[f(X)] = \int f(x)\, p(x)\, dx$ are the foundation of every statistical learning objective, from maximum likelihood to policy gradient. The Riemann integral provides the first rigorous framework for these integrals, and the *fundamental theorem of calculus* connects differentiation and integration, enabling the computation of integrals via antiderivatives. The tools developed in this lesson will let us make all of these arguments precise.

Why RL/ML needs differentiation and integration

- **Backpropagation.** The chain rule computes $\partial L / \partial \theta_i$ by composing derivatives through layers of a neural network, making gradient-based training possible.
- **Convergence of gradient descent.** The mean value theorem and Taylor's theorem establish that gradient descent with step size η decreases the loss by approximately $\eta \|\nabla L\|^2$, yielding convergence rates.
- **Policy gradient.** The REINFORCE estimator $\nabla_\theta J(\theta) = \mathbb{E}[\nabla_\theta \log \pi_\theta(a|s) \cdot Q^\pi(s,a)]$ requires exchanging differentiation and integration, justified by the dominated convergence theorem (which extends the Riemann integral framework developed here).
- **Limitations of Riemann integration.** Many functions arising in probability (e.g., indicator functions of complicated sets) are not Riemann integrable, motivating the Lebesgue integral in Phase 01.

Prerequisites. Continuity (ε-δ and sequential characterizations), the extreme value theorem, the intermediate value theorem, and uniform continuity (Lesson 2). Completeness of $\mathbb{R}$, convergence of sequences, and supremum/infimum (Lesson 1).

Learning objectives. After completing this lesson, you will be able to:

- **Define** the derivative and **prove** that differentiability implies continuity.
- **Derive** the sum, product, quotient, and chain rules for differentiation.
- **State** and **prove** Fermat's theorem, Rolle's theorem, and the mean value theorem.
- **Apply** Taylor's theorem with Lagrange remainder to obtain polynomial approximations and error bounds.
- **Define** the Riemann integral via partitions and upper/lower sums, and **prove** that continuous functions are integrable.
- **State** and **prove** the fundamental theorem of calculus (both parts).

2 References

- Rudin [1976, Chapters 5–6] — Rigorous treatment of differentiation and the Riemann–Stieltjes integral.
- Abbott [2015, Chapters 5–7] — Accessible development of derivatives, the mean value theorem, and Riemann integration.

- Bartle and Sherbert [2011, Chapters 6–7] — Detailed coverage of differentiation rules, Taylor's theorem, and the Riemann integral with many examples.
- Pugh [2015, Chapters 5–6] — Differentiation and integration with a geometric viewpoint.
- Folland [1999, Chapter 1] — Brief review of Riemann integration as motivation for the Lebesgue theory.

3 The Derivative

Definition 3.1 (Derivative)**.** Let $f\colon (a,b) \to \mathbb{R}$ and $c \in (a,b)$. The *derivative* of f at c is

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c},$$

provided this limit exists. If $f'(c)$ exists, we say f is *differentiable* at c. Equivalently, $f'(c) = \lim_{h \to 0}(f(c+h) - f(c))/h$.

Theorem 3.2 (Differentiability Implies Continuity)**.** *If f is differentiable at c, then f is continuous at c.*

Proof. Suppose $f'(c)$ exists. For $x \neq c$:

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c} \cdot (x - c).$$

Taking $x \to c$: the first factor converges to $f'(c)$ and the second to 0, so $f(x) - f(c) \to 0$ by the algebraic limit theorems (Lesson 1, Theorem 4.4). Hence $f(x) \to f(c)$, i.e., f is continuous at c. □

Example 3.3. The function $f(x) = |x|$ is continuous at 0 but not differentiable there. The left-hand limit $\lim_{h \to 0^-}(|h| - 0)/h = -1$ and the right-hand limit $\lim_{h \to 0^+} |h|/h = 1$ disagree, so the two-sided limit does not exist.

Example 3.4. For $f(x) = x^n$ with $n \in \mathbb{N}$, the derivative is $f'(x) = nx^{n-1}$. Indeed,

$$\frac{(c+h)^n - c^n}{h} = \frac{h\big(nc^{n-1} + \binom{n}{2}c^{n-2}h + \cdots + h^{n-1}\big)}{h} \to nc^{n-1}$$

as $h \to 0$.

4 Differentiation Rules

Theorem 4.1 (Sum and Constant Multiple Rules)**.** *If f and g are differentiable at c and $\alpha \in \mathbb{R}$, then:*

(i) $(f+g)'(c) = f'(c) + g'(c)$.
(ii) $(\alpha f)'(c) = \alpha f'(c)$.

Proof. (i) $\dfrac{(f+g)(x) - (f+g)(c)}{x - c} = \dfrac{f(x) - f(c)}{x - c} + \dfrac{g(x) - g(c)}{x - c} \to f'(c) + g'(c)$ as $x \to c$, by the algebraic limit theorem for sums.

(ii) is similar, using the constant multiple rule for limits. □

Theorem 4.2 (Product Rule)**.** *If f and g are differentiable at c, then $(fg)'(c) = f'(c)\,g(c) + f(c)\,g'(c)$.*

Proof. We write:

$$\begin{aligned}\frac{f(x)g(x)-f(c)g(c)}{x-c} &= \frac{f(x)g(x)-f(c)g(x)+f(c)g(x)-f(c)g(c)}{x-c}\\ &= \frac{f(x)-f(c)}{x-c}\,g(x)+f(c)\,\frac{g(x)-g(c)}{x-c}.\end{aligned}$$

As $x \to c$: the first term converges to $f'(c)\cdot g(c)$ (using $g(x)\to g(c)$ by Theorem 3.2), and the second to $f(c)\cdot g'(c)$. □

Theorem 4.3 (Quotient Rule)**.** *If f and g are differentiable at c and $g(c)\neq 0$, then*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)\,g(c)-f(c)\,g'(c)}{g(c)^2}.$$

Proof. It suffices to show $(1/g)'(c) = -g'(c)/g(c)^2$ and then apply the product rule to $f\cdot(1/g)$. Since $g(c)\neq 0$ and g is continuous at c, $g(x)\neq 0$ near c. Then:

$$\frac{1/(g(x))-1/(g(c))}{x-c} = \frac{g(c)-g(x)}{(x-c)\,g(x)\,g(c)} = -\frac{1}{g(x)\,g(c)}\cdot\frac{g(x)-g(c)}{x-c} \to -\frac{g'(c)}{g(c)^2}.$$ □

Theorem 4.4 (Chain Rule)**.** *Let f be differentiable at c and g be differentiable at $f(c)$. Then $g\circ f$ is differentiable at c and*

$$(g\circ f)'(c) = g'(f(c))\cdot f'(c).$$

Proof. Define the auxiliary function

$$\phi(y) = \begin{cases}\dfrac{g(y)-g(f(c))}{y-f(c)} & \text{if } y\neq f(c),\\ g'(f(c)) & \text{if } y = f(c).\end{cases}$$

Since g is differentiable at $f(c)$, ϕ is continuous at $f(c)$. For $x\neq c$ with $f(x)\neq f(c)$:

$$\frac{g(f(x))-g(f(c))}{x-c} = \phi(f(x))\cdot\frac{f(x)-f(c)}{x-c}.$$

This identity also holds when $f(x)=f(c)$ (both sides are 0). Taking $x\to c$: $f(x)\to f(c)$ (continuity from differentiability), so $\phi(f(x))\to\phi(f(c)) = g'(f(c))$, and $(f(x)-f(c))/(x-c)\to f'(c)$. Hence $(g\circ f)'(c) = g'(f(c))\cdot f'(c)$. □

The chain rule as the backbone of backpropagation

A neural network computes a composition $f = f_L\circ f_{L-1}\circ\cdots\circ f_1$, where each $f_\ell(\mathbf{x}) = \sigma(W_\ell\mathbf{x}+\mathbf{b}_\ell)$ applies an affine transformation followed by a nonlinearity σ. The chain rule gives

$$\frac{\partial L}{\partial W_\ell} = \frac{\partial L}{\partial f_L}\cdot\frac{\partial f_L}{\partial f_{L-1}}\cdots\frac{\partial f_{\ell+1}}{\partial f_\ell}\cdot\frac{\partial f_\ell}{\partial W_\ell},$$

which is computed efficiently by the backpropagation algorithm. Without the chain rule, computing gradients of deep networks would require finite-difference approximations with cost proportional to the number of parameters.

Example 4.5. Let $h(x) = (3x^2+1)^5$. Setting $f(x) = 3x^2+1$ and $g(y) = y^5$, the chain rule gives $h'(x) = 5(3x^2+1)^4\cdot 6x = 30x(3x^2+1)^4$.

5 Mean Value Theorem and Consequences

Theorem 5.1 (Fermat's Theorem)**.** *Let $f\colon (a,b) \to \mathbb{R}$ and suppose f attains a local maximum or minimum at $c \in (a,b)$. If f is differentiable at c, then $f'(c) = 0$.*

Proof. Suppose f has a local maximum at c: there exists $\delta > 0$ such that $f(x) \leq f(c)$ for $|x - c| < \delta$. For $0 < h < \delta$:

$$\frac{f(c+h) - f(c)}{h} \leq 0,$$

so $f'(c) = \lim_{h\to 0^+}(f(c+h) - f(c))/h \leq 0$. For $-\delta < h < 0$:

$$\frac{f(c+h) - f(c)}{h} \geq 0,$$

so $f'(c) = \lim_{h\to 0^-}(f(c+h) - f(c))/h \geq 0$. Together, $f'(c) = 0$. The proof for a local minimum is analogous. □

Theorem 5.2 (Rolle's Theorem)**.** *If $f\colon [a,b] \to \mathbb{R}$ is continuous on $[a,b]$, differentiable on (a,b), and $f(a) = f(b)$, then there exists $c \in (a,b)$ with $f'(c) = 0$.*

Proof. If f is constant on $[a,b]$, then $f'(c) = 0$ for every $c \in (a,b)$. Otherwise, f is not constant, so by the extreme value theorem (Lesson 2, Theorem 6.2), f attains its maximum M and minimum m on $[a,b]$. Since f is not constant and $f(a) = f(b)$, at least one extreme value is attained at some $c \in (a,b)$ (both cannot occur only at the endpoints, since $f(a) = f(b)$). By Fermat's theorem, $f'(c) = 0$. □

Mean Value Theorem

Theorem 5.3 (Mean Value Theorem (MVT))**.** *If $f\colon [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and differentiable on (a,b), then there exists $c \in (a,b)$ with*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Define the auxiliary function

$$g(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then g is continuous on $[a,b]$, differentiable on (a,b), and $g(a) = 0 = g(b)$. By Rolle's theorem, there exists $c \in (a,b)$ with $g'(c) = 0$, i.e.,

$$f'(c) - \frac{f(b) - f(a)}{b - a} = 0.$$ □

Example 5.4. If a car travels 150 km in 2 hours, the MVT guarantees there is an instant when the car's speed is exactly $150/2 = 75$ km/h, regardless of how the speed varied during the trip.

Corollary 5.5 (Monotonicity from the Derivative)**.** *Let f be continuous on $[a,b]$ and differentiable on (a,b).*

(i) If $f'(x) > 0$ for all $x \in (a,b)$, then f is strictly increasing on $[a,b]$.
(ii) If $f'(x) = 0$ for all $x \in (a,b)$, then f is constant on $[a,b]$.

Proof. (i) For $a \leq x_1 < x_2 \leq b$, the MVT gives $c \in (x_1, x_2)$ with $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) > 0$.

(ii) For any $x_1, x_2 \in [a,b]$, the MVT gives $|f(x_2) - f(x_1)| = |f'(c)||x_2 - x_1| = 0$. □

MVT in optimization convergence analysis

The mean value theorem is the workhorse of optimization convergence proofs. For gradient descent with step size η on a function f with L-Lipschitz gradient ($\|f'(x) - f'(y)\| \leq L\|x - y\|$), the MVT (generalized to $\mathbb{R}^n$) gives

$$f(x - \eta f'(x)) \leq f(x) - \eta\|f'(x)\|^2 + \frac{L\eta^2}{2}\|f'(x)\|^2.$$

Choosing $\eta = 1/L$ yields $f(x_{k+1}) \leq f(x_k) - \frac{1}{2L}\|f'(x_k)\|^2$, the standard descent lemma that drives $O(1/k)$ convergence rates for smooth convex functions.

Example 5.6. Let $f(x) = e^x$ on $[0, h]$ for $h > 0$. The MVT gives $c \in (0, h)$ with $e^h - 1 = e^c \cdot h$. Since $1 < e^c < e^h$, we obtain the bounds $h < e^h - 1 < h\, e^h$.

6 Taylor's Theorem

Theorem 6.1 (Taylor's Theorem with Lagrange Remainder)**.** *Let $f\colon [a, b] \to \mathbb{R}$ be such that $f, f', \ldots, f^{(n)}$ are continuous on $[a, b]$ and $f^{(n+1)}$ exists on (a, b). Then for every $x \in [a, b]$, there exists c between a and x such that*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k + \frac{f^{(n+1)}(c)}{(n + 1)!}(x - a)^{n+1}.$$

Proof. Define the remainder $R_n(x) = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k$. We seek c with $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - a)^{n+1}$.

Fix $x \neq a$ and define M by $R_n(x) = M(x - a)^{n+1}$, i.e., $M = R_n(x)/(x - a)^{n+1}$. Define

$$g(t) = f(t) - \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(t - a)^k - M(t - a)^{n+1}.$$

Then $g(a) = 0$ (all terms vanish at $t = a$) and $g(x) = R_n(x) - M(x - a)^{n+1} = 0$ by the choice of M.

By Rolle's theorem, there exists c_1 between a and x with $g'(c_1) = 0$. Now $g'(a) = 0$ (since the first n Taylor terms are chosen to match derivatives at a, and the $(n+1)$-th term contributes $(n + 1)M(t - a)^n$ which vanishes at a). Applying Rolle's theorem to g' on the interval between a and c_1, we get c_2 with $g''(c_2) = 0$. Continuing $n + 1$ times, we obtain $c = c_{n+1}$ between a and x with $g^{(n+1)}(c) = 0$. But $g^{(n+1)}(t) = f^{(n+1)}(t) - (n + 1)!\, M$, so $f^{(n+1)}(c) = (n + 1)!\, M$, giving $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - a)^{n+1}$. □

Example 6.2. For $f(x) = e^x$ centered at $a = 0$, $f^{(k)}(0) = 1$ for all k, so the Taylor polynomial is $\sum_{k=0}^{n} x^k/k!$ and the remainder is $R_n(x) = e^c x^{n+1}/(n + 1)!$ for some c between 0 and x. For any fixed x, $|R_n(x)| \leq e^{|x|}|x|^{n+1}/(n + 1)! \to 0$ as $n \to \infty$ (by the ratio test applied to $|x|^n/n!$), confirming $e^x = \sum_{k=0}^{\infty} x^k/k!$.

Example 6.3. The first-order Taylor expansion of a smooth function f at x_0 is $f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(c)}{2}(x - x_0)^2$. In optimization, setting $x = x_0 - \eta f'(x_0)$ and bounding $|f''(c)| \leq L$ gives

$$f(x_0 - \eta f'(x_0)) \leq f(x_0) - \eta(f'(x_0))^2 + \frac{L\eta^2}{2}(f'(x_0))^2,$$

the descent inequality used to analyze gradient descent.

Taylor approximations in optimization convergence

Taylor's theorem provides the mathematical foundation for convergence analysis of first- and second-order optimization methods. The second-order expansion $f(x+h) \approx f(x) + f'(x)h + \frac{1}{2}f''(x)h^2$ is used to:

- Derive the *descent lemma* for gradient descent with L-smooth functions.
- Analyze *Newton's method*, which uses the full quadratic model to achieve quadratic convergence near a minimizer.
- Justify *natural gradient* and *trust region* methods in policy optimization, where the policy is locally approximated by a quadratic model in the parameter space.

7 Riemann Integration

Definition 7.1 (Partition, Upper and Lower Sums)**.** Let $f\colon [a,b] \to \mathbb{R}$ be bounded.

(i) A *partition* of $[a,b]$ is a finite set $P = \{x_0, x_1, \ldots, x_n\}$ with $a = x_0 < x_1 < \cdots < x_n = b$.

(ii) On each subinterval $[x_{i-1}, x_i]$, define $m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}$ and $M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\}$.

(iii) The *lower sum* is $L(f,P) = \sum_{i=1}^n m_i(x_i - x_{i-1})$.

(iv) The *upper sum* is $U(f,P) = \sum_{i=1}^n M_i(x_i - x_{i-1})$.

Definition 7.2 (Refinement)**.** A partition Q is a *refinement* of P if $P \subseteq Q$.

Lemma 7.3 (Refinement Lemma)**.** *If Q is a refinement of P, then $L(f,P) \le L(f,Q) \le U(f,Q) \le U(f,P)$.*

Proof. It suffices to prove the result when Q has one additional point x^* in $[x_{j-1}, x_j]$; the general case follows by induction. The lower sum changes only on $[x_{j-1}, x_j]$:

$$\begin{aligned}&\inf_{[x_{j-1},x^*]} f \cdot (x^* - x_{j-1}) + \inf_{[x^*,x_j]} f \cdot (x_j - x^*)\\ &\qquad \ge \inf_{[x_{j-1},x_j]} f \cdot \big((x^* - x_{j-1}) + (x_j - x^*)\big) = m_j(x_j - x_{j-1}),\end{aligned}$$

since $\inf_{[x_{j-1},x^*]} f \ge \inf_{[x_{j-1},x_j]} f$ and similarly for $[x^*, x_j]$. Hence $L(f,Q) \ge L(f,P)$. The argument for $U(f,Q) \le U(f,P)$ is analogous (with sup and the inequality reversed). Finally, $L(f,Q) \le U(f,Q)$ since $m_i \le M_i$ on each subinterval. □

Corollary 7.4. *For any two partitions P_1, P_2 of $[a,b]$: $L(f,P_1) \le U(f,P_2)$.*

Proof. Let $Q = P_1 \cup P_2$ (a common refinement). Then $L(f,P_1) \le L(f,Q) \le U(f,Q) \le U(f,P_2)$ by Lemma 7.3. □

Definition 7.5 (Riemann Integral)**.** Let $f\colon [a,b] \to \mathbb{R}$ be bounded. Define the *lower integral* and *upper integral*:

$$\underline{\int_a^b} f = \sup_P L(f,P), \qquad \overline{\int_a^b} f = \inf_P U(f,P),$$

where the supremum and infimum are over all partitions P of $[a,b]$. By Corollary 7.4, $\underline{\int_a^b} f \le \overline{\int_a^b} f$. We say f is *Riemann integrable* on $[a,b]$ if $\underline{\int_a^b} f = \overline{\int_a^b} f$, and the common value is denoted $\int_a^b f(x)\,dx$ or $\int_a^b f$.

Theorem 7.6 (Integrability Criterion)**.** *A bounded function f on $[a,b]$ is Riemann integrable if and only if for every $\varepsilon > 0$ there exists a partition P such that $U(f,P) - L(f,P) < \varepsilon$.*

Proof. ($\Rightarrow$) Suppose f is integrable with $\underline{\int} f = \overline{\int} f = I$. Given $\varepsilon > 0$, by definition of supremum and infimum, choose P_1 with $L(f, P_1) > I - \varepsilon/2$ and P_2 with $U(f, P_2) < I + \varepsilon/2$. Let $P = P_1 \cup P_2$. Then $L(f, P) \geq L(f, P_1) > I - \varepsilon/2$ and $U(f, P) \leq U(f, P_2) < I + \varepsilon/2$, so $U(f, P) - L(f, P) < \varepsilon$.

($\Leftarrow$) For each n, choose P_n with $U(f, P_n) - L(f, P_n) < 1/n$. Then $\overline{\int} f - \underline{\int} f \leq U(f, P_n) - L(f, P_n) < 1/n$ for all n. Hence $\overline{\int} f = \underline{\int} f$. □

Theorem 7.7 (Continuous Functions Are Integrable)**.** *If $f\colon [a, b] \to \mathbb{R}$ is continuous, then f is Riemann integrable.*

Proof. Since $[a, b]$ is compact and f is continuous, f is uniformly continuous (Lesson 2, Theorem 7.4). Given $\varepsilon > 0$, choose $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon/(b - a)$ whenever $|x - y| < \delta$. Take any partition $P = \{x_0, \ldots, x_n\}$ with mesh $\|P\| = \max_i(x_i - x_{i-1}) < \delta$. Then on each $[x_{i-1}, x_i]$, $M_i - m_i < \varepsilon/(b - a)$ (since the subinterval has diameter less than δ), so

$$U(f, P) - L(f, P) = \sum_{i=1}^{n} (M_i - m_i)(x_i - x_{i-1}) < \frac{\varepsilon}{b - a} \sum_{i=1}^{n} (x_i - x_{i-1}) = \varepsilon.$$

By Theorem 7.6, f is integrable. □

Example 7.8. Let $f(x) = x^2$ on $[0, 1]$. Using the uniform partition $P_n = \{0, 1/n, 2/n, \ldots, 1\}$:

$$L(f, P_n) = \sum_{i=1}^{n} \left(\frac{i-1}{n}\right)^2 \cdot \frac{1}{n} = \frac{1}{n^3} \sum_{i=0}^{n-1} i^2 = \frac{(n-1)n(2n-1)}{6n^3} \to \frac{1}{3}.$$

Similarly, $U(f, P_n) = \frac{n(n+1)(2n+1)}{6n^3} \to 1/3$. Since $U(f, P_n) - L(f, P_n) \to 0$, we confirm $\int_0^1 x^2\, dx = 1/3$.

8 Properties of the Integral

Theorem 8.1 (Properties of the Riemann Integral)**.** *Let f, g be Riemann integrable on $[a, b]$ and $\alpha \in \mathbb{R}$. Then:*

(i) ***Linearity:*** $\int_a^b (\alpha f + g) = \alpha \int_a^b f + \int_a^b g$.

(ii) ***Monotonicity:*** *If $f(x) \leq g(x)$ for all $x \in [a, b]$, then $\int_a^b f \leq \int_a^b g$.*

(iii) ***Additivity:*** *For $a < c < b$, $\int_a^b f = \int_a^c f + \int_c^b f$.*

(iv) ***Triangle inequality:*** $\left|\int_a^b f\right| \leq \int_a^b |f|$.

Proof. (i) For the scalar multiple: on each subinterval, $\sup(\alpha f) = \alpha \sup f$ if $\alpha \geq 0$ (and $\alpha \inf f$ if $\alpha < 0$), so $U(\alpha f, P) = \alpha\, U(f, P)$ for $\alpha \geq 0$. The case $\alpha < 0$ uses $U(\alpha f, P) = \alpha\, L(f, P)$. Taking suprema/infima over partitions gives $\int \alpha f = \alpha \int f$.

For the sum: on each subinterval $[x_{i-1}, x_i]$, $\inf f + \inf g \leq \inf(f + g) \leq \sup(f + g) \leq \sup f + \sup g$, so $L(f, P) + L(g, P) \leq L(f + g, P)$ and $U(f + g, P) \leq U(f, P) + U(g, P)$. Given $\varepsilon > 0$, choose P_1, P_2 with $U(f, P_1) - L(f, P_1) < \varepsilon/2$ and $U(g, P_2) - L(g, P_2) < \varepsilon/2$. Let $P = P_1 \cup P_2$. Then

$$U(f + g, P) - L(f + g, P) \leq \big(U(f, P) - L(f, P)\big) + \big(U(g, P) - L(g, P)\big) < \varepsilon,$$

so $f + g$ is integrable. The value equals $\int f + \int g$ by taking limits of upper sums.

(ii) If $f \leq g$, then for any partition P, $U(f, P) \leq U(g, P)$, so $\int f = \inf_P U(f, P) \leq \inf_P U(g, P) = \int g$.

(iii) Given $\varepsilon > 0$, choose partitions P' of $[a, c]$ and P'' of $[c, b]$ with $U(f, P') - L(f, P') < \varepsilon/2$ and $U(f, P'') - L(f, P'') < \varepsilon/2$. Then $P = P' \cup P''$ is a partition of $[a, b]$ with $L(f, P) = L(f, P') + L(f, P'')$ and $U(f, P) = U(f, P') + U(f, P'')$. The result follows by taking $\varepsilon \to 0$.

(iv) Since $-|f(x)| \leq f(x) \leq |f(x)|$, monotonicity gives $-\int |f| \leq \int f \leq \int |f|$. □

9 Fundamental Theorem of Calculus

Fundamental Theorem of Calculus

Theorem 9.1 (FTC Part I). *Let $f\colon [a,b] \to \mathbb{R}$ be Riemann integrable. Define $F(x) = \int_a^x f(t)\,dt$ for $x \in [a,b]$. Then:*
(i) F is continuous on $[a,b]$.
(ii) If f is continuous at $c \in (a,b)$, then F is differentiable at c and $F'(c) = f(c)$.

Theorem 9.2 (FTC Part II). *If $f\colon [a,b] \to \mathbb{R}$ is continuous and G is any antiderivative of f (i.e., $G'(x) = f(x)$ for all $x \in (a,b)$, with G continuous on $[a,b]$), then*

$$\int_a^b f(x)\,dx = G(b) - G(a).$$

Proof of FTC Part I. (i) Since f is bounded, say $|f(t)| \le M$ for $t \in [a,b]$, we have for $x, y \in [a,b]$ with $x < y$:

$$|F(y) - F(x)| = \left|\int_x^y f(t)\,dt\right| \le \int_x^y |f(t)|\,dt \le M(y-x).$$

Given $\varepsilon > 0$, taking $\delta = \varepsilon/M$ gives $|F(y) - F(x)| < \varepsilon$ whenever $|y - x| < \delta$. Hence F is (uniformly) continuous.

(ii) Suppose f is continuous at c. Given $\varepsilon > 0$, choose $\delta > 0$ with $|f(t) - f(c)| < \varepsilon$ for $|t - c| < \delta$. For $0 < |h| < \delta$:

$$\frac{F(c+h) - F(c)}{h} - f(c) = \frac{1}{h}\int_c^{c+h} f(t)\,dt - f(c) = \frac{1}{h}\int_c^{c+h} \big(f(t) - f(c)\big)\,dt.$$

For $h > 0$: $\left|\frac{1}{h}\int_c^{c+h}(f(t) - f(c))\,dt\right| \le \frac{1}{h}\int_c^{c+h}|f(t) - f(c)|\,dt < \frac{1}{h}\cdot \varepsilon h = \varepsilon$. The case $h < 0$ is analogous. Hence $F'(c) = f(c)$. □

Proof of FTC Part II. Let $F(x) = \int_a^x f(t)\,dt$. By Part I, $F'(x) = f(x) = G'(x)$ for $x \in (a,b)$. Hence $(F-G)'(x) = 0$ on (a,b), so $F-G$ is constant on $[a,b]$ by Corollary 5.5(ii). Since $F(a) = 0$, we get $F(x) = G(x) - G(a)$ for all $x \in [a,b]$. Evaluating at $x = b$: $\int_a^b f = F(b) = G(b) - G(a)$. □

Example 9.3. To compute $\int_0^1 x^3\,dx$: an antiderivative of $f(x) = x^3$ is $G(x) = x^4/4$. By FTC Part II, $\int_0^1 x^3\,dx = G(1) - G(0) = 1/4 - 0 = 1/4$.

Example 9.4. Let $F(x) = \int_0^x e^{-t^2}\,dt$. By FTC Part I, $F'(x) = e^{-x^2}$. While F cannot be expressed in closed form using elementary functions, the FTC guarantees its differentiability and gives the derivative explicitly. The function F is closely related to the error function $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}}F(x)$, ubiquitous in probability and statistics.

Limitations of the Riemann integral

The Riemann integral has fundamental limitations:

(i) The indicator function $\mathbf{1}_{\mathbb{Q}}$ of the rationals on $[0,1]$ is not Riemann integrable: $L(f,P) = 0$ and $U(f,P) = 1$ for every partition P (since every subinterval contains both rationals and irrationals).

(ii) Pointwise limits of Riemann integrable functions need not be Riemann integrable, and even when they are, the limit of the integrals need not equal the integral of the limit.

(iii) The Riemann integral does not interact well with measure theory: it cannot integrate with respect to general measures.

These limitations motivate the *Lebesgue integral* (Phase 01, Measure Theory), which resolves all three issues and provides the foundation for modern probability theory.

10 Exercises

Exercise 10.1 (Differentiation from the definition). $(*)$
Using Definition 3.1, prove that $f(x) = 1/(x+1)$ is differentiable on $(-1, \infty)$ and compute $f'(x)$.

Exercise 10.2 (Chain rule application). $(*)$
Let $g(x) = \sin(x^2 + 1)$. Compute $g'(x)$ using the chain rule (Theorem 4.4), identifying the inner and outer functions explicitly.

Exercise 10.3 (MVT in gradient descent). $(**)$
Let $f\colon \mathbb{R} \to \mathbb{R}$ be differentiable with $|f'(x) - f'(y)| \le L|x - y|$ for all $x, y \in \mathbb{R}$ (i.e., f' is Lipschitz with constant L).

(a) Use Taylor's theorem (Theorem 6.1 with $n = 1$) to show that for all $x, h \in \mathbb{R}$: $|f(x+h) - f(x) - f'(x)h| \le \frac{L}{2}h^2$.

(b) Deduce that the gradient descent update $x_{k+1} = x_k - \frac{1}{L}f'(x_k)$ satisfies $f(x_{k+1}) \le f(x_k) - \frac{1}{2L}(f'(x_k))^2$.

Hint: For (a), the Lagrange remainder involves $f''(c)$, and $|f''(c)| \le L$ follows from the Lipschitz condition on f' via the MVT.

Exercise 10.4 (Integrability of monotone functions). $(**)$
Prove that every monotone function $f\colon [a, b] \to \mathbb{R}$ is Riemann integrable.

Hint: Assume f is increasing. Use the uniform partition P_n with n equal subintervals, and show $U(f, P_n) - L(f, P_n) = \frac{b-a}{n}(f(b) - f(a)) \to 0$.

Exercise 10.5 (FTC and the Gaussian integral). $(**)$
Define $F(x) = \int_0^x e^{-t^2/2}\, dt$.

(a) Use FTC Part I to show F is differentiable and find $F'(x)$.

(b) Show F is strictly increasing on $[0, \infty)$ and bounded above. Conclude that $\lim_{x\to\infty} F(x)$ exists.

(c) Use the bound $e^{-t^2/2} \le e^{-t/2}$ for $t \ge 1$ to show $F(x) \le 1 + 2e^{-1/2}$ for all $x \ge 0$.

Exercise 10.6 (Non-integrability and measure theory motivation). $(***)$
Let $f = \mathbf{1}_{\mathbb{Q}}$ on $[0, 1]$ (i.e., $f(x) = 1$ if $x \in \mathbb{Q}$, $f(x) = 0$ otherwise).

(a) Prove that $L(f, P) = 0$ and $U(f, P) = 1$ for every partition P of $[0, 1]$.

(b) Conclude that f is not Riemann integrable.

(c) The Lebesgue integral assigns $\int_{[0,1]} f\, d\lambda = 0$ (since $\mathbb{Q}$ has Lebesgue measure zero). Explain informally why this is the "right" answer and why the Riemann integral fails to capture it.

11 Summary

- The *derivative* $f'(c) = \lim_{h\to 0}(f(c + h) - f(c))/h$ measures the instantaneous rate of change; differentiability at c implies continuity at c, but not conversely (e.g., $|x|$ at 0).
- The *sum, product, quotient, and chain rules* reduce differentiation of complex expressions to differentiation of simpler components. The chain rule is the mathematical basis of backpropagation.
- *Fermat's theorem*: at an interior extremum of a differentiable function, the derivative vanishes. This leads to *Rolle's theorem* and the *mean value theorem*, which connects the derivative to the function's total change.

- *Taylor's theorem* with Lagrange remainder provides polynomial approximations with explicit error bounds: $|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$ where $M = \sup |f^{(n+1)}|$. This underpins convergence analysis of gradient descent and Newton's method.
- The *Riemann integral* is defined via upper and lower sums over partitions. A bounded function is integrable if and only if for every $\varepsilon > 0$ there exists a partition with $U(f,P) - L(f,P) < \varepsilon$.
- *Continuous functions are Riemann integrable*, via uniform continuity on compact sets.
- *FTC Part I*: the integral $F(x) = \int_a^x f$ is continuous, and $F'(c) = f(c)$ at points of continuity of f. *FTC Part II*: $\int_a^b f = G(b) - G(a)$ for any antiderivative G.
- The Riemann integral has fundamental limitations (non-integrable functions, poor behavior under limits) that motivate the *Lebesgue integral* developed in Phase 01.

12 Further Reading

- **Lesson 4** (Sequences and Series of Functions) extends differentiation and integration to function sequences, addressing when differentiation and integration commute with limits—a question whose definitive answer comes from the Lebesgue theory.
- In **Phase 01** (Measure Theory, Lessons 4–5), the Riemann integral is replaced by the Lebesgue integral, which integrates a much larger class of functions and satisfies powerful limit theorems (monotone convergence, dominated convergence) that the Riemann integral lacks.
- In **Phase 01** (Optimization, Lessons 8–10), the mean value theorem and Taylor's theorem are extended to $\mathbb{R}^n$ and used systematically to analyze gradient descent, proximal methods, and interior-point methods.
- For multivariable differentiation (partial derivatives, the Jacobian, and the general chain rule), see Rudin [1976, Chapter 9] or Pugh [2015, Chapter 5]. These are essential for understanding backpropagation in neural networks with vector-valued layers.
- Bartle and Sherbert [2011, Chapter 6] provides an extensive collection of examples and counterexamples for Riemann integrability, clarifying the boundary between integrable and non-integrable functions.
- For historical context, the development of the integral from Cauchy through Riemann to Lebesgue is discussed in Folland [1999, Chapter 1], providing motivation for the modern measure-theoretic approach.

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